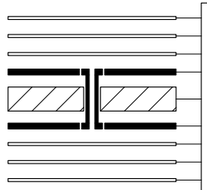
LAYER STACKUP LEGEND



Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**L1 (Sig, PWR)**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Core		1.49mm	FR4_7628	Dielectric	
Copper	**L2 (Sig, GND)**	**0.035mm (1.00oz)**		**Plane**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.6mm
Note: external layer thicknesses are specified after plating

IMPEDANCE TABLE

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: IMMERSION GOLD

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TO BE NON-TENTED.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

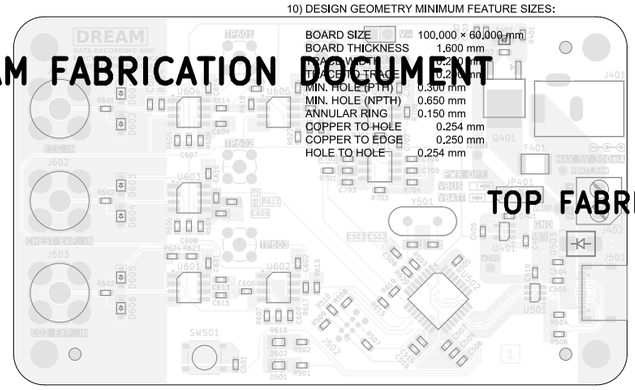
9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY RYAN DYNAMICS.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	100.000 × 60.000 mm
BOARD THICKNESS	1.600 mm
[illegible]	[illegible]
MIN. HOLE (PTH)	0.300 mm
MIN. HOLE (NPTH)	0.650 mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.254 mm
COPPER TO EDGE	0.250 mm
HOLE TO HOLE	0.254 mm

DREAM FABRICATION DOCUMENT



TOP FABRICATION (SCALE 1:1)

ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

PROJECT MENG420 PROJ	Ryan Dynamics	
DRN 2024-04-13 R. HICKS	THIS DESIGN AND/OR DRAWING IS THE PROPERTY OF RYAN DYNAMICS AND SHALL NOTBE REPRODUCED WITHOUT AUTHORISATION.	
CHK		
ENG APP	DREAM	
MFR APP		
SHEET PATH	GIT HASH 45c44fe	DRAWING No FABRICATION
FILENAME DREAM.kicad_pcb	VARIANT PRELIMINARY	REVISION + (Unreleased) SHEET 1 OF 8

A3

DREAM FABRICATION DOCUMENT

BOTTOM FABRICATION (SCALE 1:1)

ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

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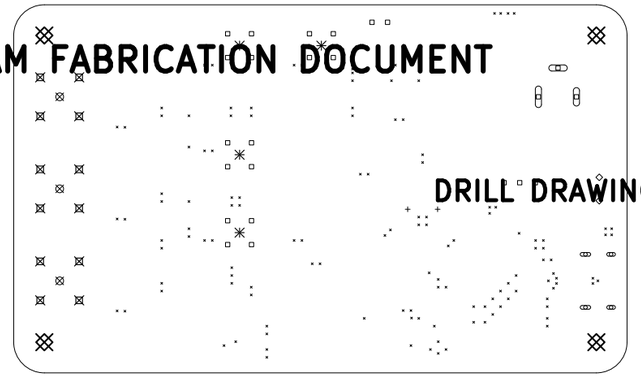
CHK			
ENG APP	DREAM		
MFR APP			
SHEET PATH	GIT HASH 45c44fe	DRAWING No FABRICATION	A3
FILENAME DREAM.kicad_pcb	VARIANT PRELIMINARY	REVISION + (Unreleased)	SHEET 2 OF 8

DRILL TABLE

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	131	0,30mm (11,81mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Via
○	4	0,60mm (23,62mils)	PTH	Slot	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
+	2	0,80mm (31,50mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
◻	21 + 3	1,00mm (39,37mils)	PTH	Round + Slot	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
◇	2	1,10mm (43,31mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⊠	3	1,27mm (50,00mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
✳	4	1,50mm (59,06mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⊠	12	1,52mm (60,00mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⊠	4	2,70mm (106,30mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
	Total 186					

DREAM FABRICATION DOCUMENT



DRILL DRAWING L1 – L2 (SCALE 1:1)

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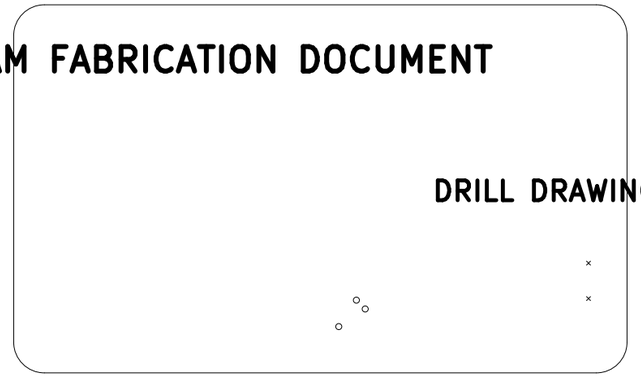
CHK			
ENG APP	DREAM		
MFR APP			
SHEET PATH	GIT HASH 45c44fe	DRAWING No FABRICATION	A3
FILENAME DREAM.kicad_pcb	VARIANT PRELIMINARY	REVISION + (Unreleased)	SHEET 3 OF 8

DRILL TABLE

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	0,65mm (25,59mils)	NPTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Mechanical
○	3	0,99mm (39,00mils)	NPTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Mechanical
	Total: 5					

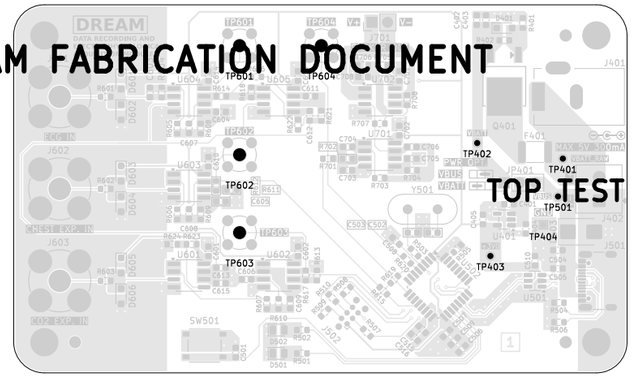
DREAM FABRICATION DOCUMENT

DRILL DRAWING L1 – L2 (SCALE 1:1)



PROJECT MENG420 PROJ	Ryan Dynamics THIS DESIGN AND/OR DRAWING IS THE PROPERTY OF RYAN DYNAMICS AND SHALL NOTBE REPRODUCED WITHOUT AUTHORISATION.		
DRN 2024-04-13 R. HICKS			
CHK			
ENG APP	DREAM		
MFR APP			
SHEET PATH	GIT HASH 45c44fe	DRAWING No FABRICATION	A3
FILENAME DREAM.kicad_pcb	VARIANT PRELIMINARY	REVISION + (Unreleased)	SHEET 4 OF 8

DREAM FABRICATION DOCUMENT



TOP TEST POINTS (SCALE 1:1)

Ref.	Net	X [mm]	Y [mm]
TP401	VBATT_RAW	89.53	34.92
TP402	VBATT	75.56	37.46
TP403	+3V0	77.72	19.05
TP404	GND	86.36	24.13
TP501	VBUS	88.75	28.75
TP601	ECG_FILT	36.83	53.34
TP602	CE_FILT	36.83	35.56
TP603	CO2_OUT	36.83	22.86
TP604	ECG_OUT	50.16	53.34

Ref.	Net	X [mm]	Y [mm]

ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

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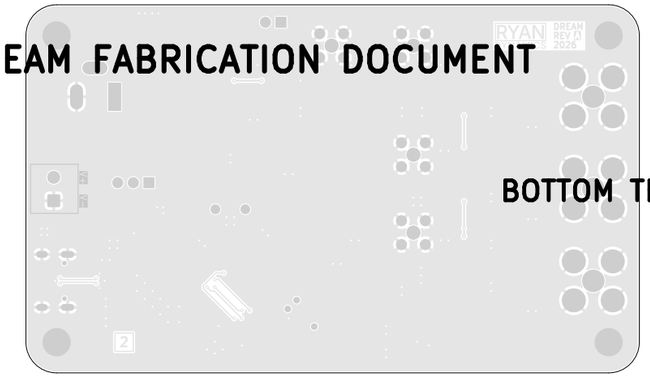
CHK				
ENG APP	DREAM			
MFR APP				
SHEET PATH	GIT HASH 45c44fe	DRAWING No FABRICATION		A3
FILENAME DREAM.kicad_pcb	VARIANT PRELIMINARY	REVISION + (Unreleased)	SHEET 5 OF 8	

Ref.	Net	X [mm]	Y [mm]

DREAM FABRICATION DOCUMENT



Y
X

BOTTOM TEST POINTS (SCALE 1:1)

ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

PROJECT MENG420 PROJ	Ryan Dynamics		
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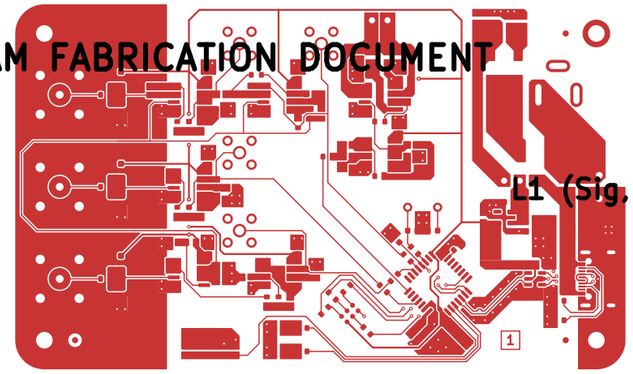
CHK			
ENG APP	DREAM		
MFR APP			
SHEET PATH	GIT HASH 45c44fe	DRAWING No FABRICATION	A3
FILENAME DREAM.kicad_pcb	VARIANT PRELIMINARY	REVISION + (Unreleased)	SHEET 6 OF 8

DREAM FABRICATION DOCUMENT



L1 (Sig. PWR) (SCALE 1:1)

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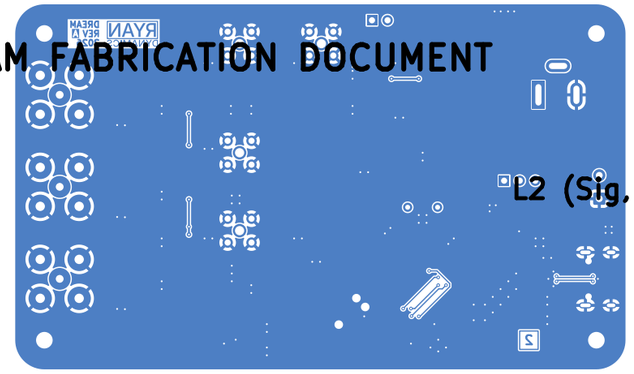
DRN 2024-04-13 R. HICKS			
CHK			
ENG APP	DREAM		
MFR APP			
SHEET PATH	GIT HASH 45c44fe	DRAWING No FABRICATION	A3
FILENAME DREAM.kicad_pcb	VARIANT PRELIMINARY	REVISION + (Unreleased)	SHEET 7 OF 8

DREAM FABRICATION DOCUMENT



L2 (Sig, GND) (SCALE 1:1)

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DRN 2024-04-13 R. HICKS			
CHK			
ENG APP	DREAM		
MFR APP			
SHEET PATH	GIT HASH 45c44fe	DRAWING No FABRICATION	A3
FILENAME DREAM.kicad_pcb	VARIANT PRELIMINARY	REVISION + (Unreleased)	SHEET 8 OF 8